FUNDAMENTALS IN AI AND ML

PROJECT REPORT

CANTEEN DEMAND PREDICTOR

Submitted By	Anoushka Gautam
Registration Number	25BHI10007
Faculty	Suneet Joshi
Slot	A14+D11+D12
Course code	CSA2001
Course Name	Fundamentals in AI and ML
Semester	Winter Semester 2025-26

Description

Food management in a college canteen is challenging due to varying daily demand. Over-preparation causes food wastage, while under-preparation leads to shortages.

This project predicts daily canteen food demand using a simple machine learning approach. The prediction is based on factors such as the day of the week, exam days, weather conditions, previous day's demand, and special events.

Python and a basic Linear Regression model are used to make the prediction.

Features

- Predicts daily food demand numerically
- Beginner-friendly Python implementation
- Uses CSV file for data storage
- Displays actual vs predicted demand
- Provides graphical visualization of results

Objectives

1. To understand the basic workflow of machine learning
2. To learn data handling using CSV files
3. To apply encoding for categorical data
4. To use Linear Regression for prediction
5. To visualize prediction results using graphs

Algorithm Used

Linear Regression is a supervised machine learning algorithm used to predict numerical values.

In this project, it learns the relationship between input factors and food demand using past data and predicts future demand.

It is chosen because it is simple, easy to understand, and suitable for beginners.

Additional Information

- **Language**: Python
- **Libraries**: Pandas, Matplotlib, Scikit-learn
- **Dataset**: CSV file containing day, weather, exam status, previous demand, and actual demand
- **Preprocessing**: Encoding text data, handling missing values, train-test split
- **Output**: Printed predictions and a demand comparison graph

Conclusion

This project shows how machine learning can be used to solve a practical problem like canteen demand prediction.

It provides hands-on experience with data preprocessing, model training, and result visualization using basic Python skills.

The project builds a strong foundation for learning more advanced machine learning concepts in the future.